

SaaS Product Engagement & Expansion Intelligence

Project Type: Data Analytics Project

Domain: Software-as-a-Service (SaaS)

Tools: Excel, SQL, Python, Pandas, Jupyter Notebook, Power BI, Git, GitHub

1. Project Overview

SaaS Product Engagement & Expansion Intelligence is a comprehensive data analytics project designed to analyze customer subscriptions, product engagement, and business performance within a Software-as-a-Service environment.

The project focuses on transforming raw customer and operational data into meaningful business information through data cleaning, SQL analysis, Python-based exploratory and statistical analysis, and Power BI visualization.

The final solution provides an interactive dashboard that allows business users to monitor important KPIs, analyze subscription trends, understand customer segments, evaluate feature adoption, review support issues, and identify business opportunities.

The project follows a complete analytics workflow from data collection and validation to analysis, visualization, insights, recommendations, and business reporting.

2. Problem Statement

SaaS businesses generate large amounts of customer, subscription, payment, product usage, and operational data. Without proper analysis, it becomes difficult for business users to understand overall performance, identify customer patterns, monitor subscription activity, and evaluate pricing and engagement behavior.

The project addresses this challenge by organizing and analyzing SaaS customer and operational data and presenting the results through an interactive Power BI dashboard.

The solution provides a centralized view of business performance and enables users to analyze the data from different dimensions such as plan category, region, subscription period, customer segment, company size, customer stage, and support category.

3. Project Objectives

The main objectives of the project are:

- Analyze subscription performance and customer engagement.
- Monitor important business KPIs including revenue, subscriptions, and customer growth.
- Identify customer adoption patterns across regions and plan categories.
- Evaluate pricing strategy using discount and payment analysis.
- Analyze customer segmentation and company-size patterns.
- Understand feature adoption across customer stages.
- Analyze support issues by category.
- Perform SQL-based business analysis.
- Perform Python-based exploratory and statistical analysis.
- Develop reusable business KPIs.
- Build an interactive Power BI dashboard.
- Convert analytical findings into business insights and recommendations

4. Project Scope

In Scope

- Customer and operational data analysis.
- Data profiling and quality assessment.
- Data cleaning and validation.
- Excel-based initial analysis.
- SQL-based data processing and KPI analysis.
- Python/Pandas exploratory analysis.
- Statistical analysis.
- Power BI data modeling and visualization.
- Business insights and recommendations.
- Dashboard documentation.
- Final project documentation and presentation.

Project Outcome

The project produces a complete business intelligence solution that converts raw SaaS data into interactive analysis and decision-support information.

5. Dataset Details

Dataset Name : SaaS Customer Journey & Operational Insights Dataset

Dataset Files

- `product_operations.csv`
- `customer_profile.csv`

Dataset Size

- Product Operations: **20,000 records**
- Customer Profiles: **5,000 records**

Primary Keys

- `record_id`
- `customer_id`

The datasets contain information required to analyze customer profiles, subscription activity, payment behavior, product operations, customer segmentation, and other business-related metrics.

6. Technology Stack

The project was developed using the following technologies:

Microsoft Excel

Used for:

- Initial data inspection
- Filtering
- Conditional formatting
- Data cleaning
- Pivot-table analysis
- Initial business summaries

MySQL Workbench 8.0

Used for:

- Data cleaning
- Joins
- Aggregations
- KPI queries

- SQL views
- Window functions
- Business analysis

Python

Used for:

- Data cleaning
- Exploratory Data Analysis
- Statistical analysis
- Outlier detection
- Correlation analysis
- Variance analysis

Pandas

Used for:

- Data manipulation
- Data transformation
- Missing-value handling
- Grouping and aggregation
- Data validation

Jupyter Notebook

Used for documenting and executing Python-based analysis.

Power BI

Used for:

- Data modeling
- DAX measures
- KPI cards
- Interactive visualizations
- Slicers
- Drill-down analysis
- Dashboard storytelling

Git & GitHub

Used for:

- Version control

- Repository management
- Project file organization
- Code and documentation storage

7. Project Repository Structure

SaaS-Product-Engagement-Expansion-Intelligence/

```
├── data/
├── docs/
├── sql/
├── notebooks/
├── dashboard/
└── README.md
```

Folder Purpose

data/

Contains project datasets and prepared data files.

docs/

Contains project documentation, reports, data dictionary, sprint documentation, and supporting documents.

sql/

Contains SQL scripts, schema definitions, queries, views, and SQL documentation.

notebooks/

Contains Python/Jupyter notebooks used for cleaning, EDA, and statistical analysis.

dashboard/

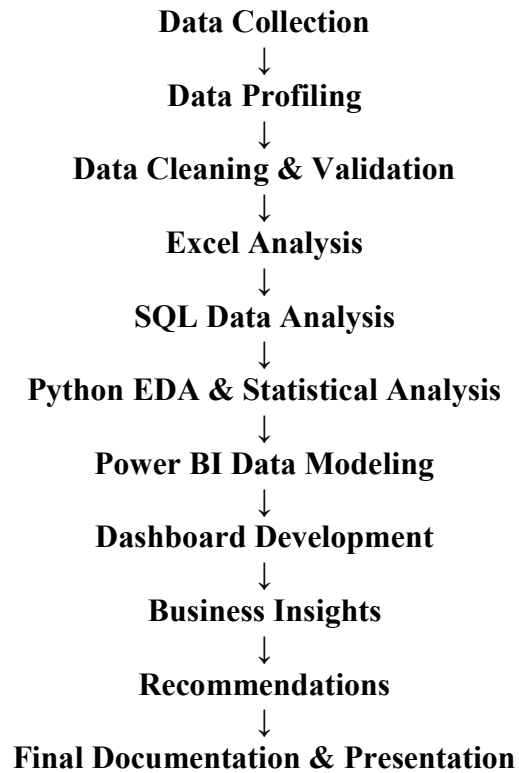
Contains Power BI dashboard files and related dashboard resources.

README.md

Provides an overview of the project, objectives, dataset, tools, workflow, dashboard, and key insights.

8. Project Workflow

The project followed the following end-to-end workflow:



9. Data Analysis & Methodology

The methodology was divided into multiple stages.

Stage 1 — Data Understanding

The datasets were first reviewed to understand their structure, columns, records, business meaning, and relationships.

A data dictionary was prepared to map technical column names to business meanings.

Stage 2 — Data Profiling

The datasets were checked for:

- Row and column counts
- Data types

- Missing values
- Duplicate records
- Invalid values
- Inconsistent categories
- Date-format issues

Stage 3 — Data Cleaning

The cleaning process included:

- Correcting data types
- Handling missing values
- Removing or identifying duplicate records
- Trimming unnecessary whitespace
- Harmonizing categories
- Validating numerical fields
- Correcting subscription date formats

Stage 4 — SQL Analysis

SQL was used to reproduce cleaning logic and perform structured analysis using:

- Joins
- Aggregations
- Grouping
- KPI calculations
- SQL views
- Window functions
- Ranking
- Running totals
- LAG
- LEAD

Stage 5 — Python Analysis

Python and Pandas were used for:

- Exploratory Data Analysis
- Summary statistics
- Distribution analysis
- Outlier detection
- Correlation analysis
- Variance analysis
- Date-based analysis
- Customer segmentation

Stage 6 — Power BI

The processed data was imported into Power BI.

DAX measures were created for important business metrics and interactive dashboard pages were developed.

10. Data Cleaning & Validation

Data quality was assessed before performing business analysis.

The major cleaning activities included:

- Identifying missing values.
- Detecting duplicate records.
- Correcting incorrect data types.
- Handling malformed subscription dates.
- Checking numerical fields for invalid values.
- Trimming unnecessary whitespace.
- Standardizing `plan_category`.
- Standardizing `payment_status`.
- Standardizing `subscription_status`.
- Validating row counts after cleaning.

The cleaning process was implemented first in Excel and subsequently reproduced using SQL and Python to ensure consistency.

11. Excel Analysis

Excel was used during the initial stages of the project for data inspection and business analysis.

Pivot tables were created using:

- Plan Category
- Payment Status
- Subscription Status

The main numerical metrics analyzed were:

- Discount Percentage
- Tax Amount
- Final Amount Paid

12. SQL Analysis

SQL was used to perform structured data analysis after loading the datasets into the database.

The major SQL activities included:

- Creating database tables.
- Defining keys.
- Reproducing cleaning operations.
- Joining customer and operational data.
- Aggregating business metrics.
- Creating KPI queries.
- Creating SQL views.
- Applying window functions.

Window functions were used for analytical operations such as:

- Ranking
- Running totals
- Previous-period comparison
- Next-period comparison

SQL analysis provided reusable and structured outputs for further analytical work.

13. Python Analysis

Python with Pandas was used for detailed exploratory and statistical analysis.

The analysis included:

Descriptive Analysis

- Mean
- Median
- Minimum
- Maximum
- Standard deviation
- Other summary statistics

Distribution Analysis

- Histograms
- Skewness
- Kurtosis

Outlier Analysis

Outliers were identified using:

- IQR method
- Z-score method

Correlation Analysis

A correlation matrix was created to examine relationships between:

- `discount_percent`
- `tax_amount`
- `final_amount_paid`

Variance Analysis

Variance analysis was performed across:

- Plan Category
- Payment Status
- Subscription Status

These analyses helped validate patterns and support the business interpretation of the dataset.

14. Power BI Dashboard

The final dashboard was developed in Power BI to provide an interactive business intelligence experience.

The dashboard contains four main sections:

1. Overview
2. Subscription Trends
3. Customer Segments
4. Business Insights

Overview Page

The **Overview** page provides a high-level summary of the SaaS business.

KPI Cards

- Total Revenue
- Total Customers

- Total Subscriptions
- Average Payment
- Average Discount
- Total Tax

Additional Visuals

- Monthly Average Payment Trend
- Revenue by Billing Cycle
- Monthly Feature Usage
- Customer Satisfaction by Support Team

The Overview page is designed for management users who need a quick understanding of overall business performance.

Subscription Trends Page

The **Subscription Trends** page focuses on time-based subscription and financial analysis.

Visuals

Total Subscriptions Growth

Shows how subscription activity changes over time.

Average Revenue Trend

Shows changes in average payment across subscription months.

Total Subscriptions by Subscription Year

Compares subscription distribution across available years.

Monthly Tax Collection

Shows changes in tax collection across different months.

This page helps users understand subscription growth and financial patterns over time.

Customer Segments Page

The **Customer Segments** page focuses on customer composition and behavior.

Visuals

Customer by Segment

Shows customer distribution across:

- Startup
- SMB
- Mid-Market
- Enterprise

Average Company Revenue by Company Size

Compares average company revenue across company-size categories.

Support Issues by Category

Shows support issues across categories such as:

- Billing
- Feature Requests
- Integration
- Login
- Technical

Feature Adoption by Customer Stage

Compares feature adoption across:

- New
- Adopted
- Power User

This page helps businesses understand customer composition, revenue patterns, support requirements, and feature adoption.

Business Insights & Recommendations

The dashboard analysis resulted in several business insights.

Insight 1 — Premium Plan Performance

Premium subscription plans demonstrate stronger customer adoption compared with standard plans.

Recommendation

The business can strengthen premium-plan positioning, enhance premium features, and use targeted upselling strategies to increase premium adoption.

Insight 2 — Subscription Growth Pattern

Subscription activity shows fluctuations across different periods.

Recommendation

The business can plan promotional activities and customer engagement campaigns during slower periods to support consistent subscription growth.

Insight 3 — Regional Opportunity

Customer adoption varies across different regions.

Recommendation

Regional performance can be analyzed further and targeted marketing and sales strategies can be developed for high-potential regions.

Insight 4 — Pricing Strategy

The current pricing and discount approach supports customer acquisition while maintaining overall business stability.

Recommendation

Discount effectiveness should be monitored regularly to improve customer acquisition without unnecessarily reducing revenue.

Insight 5 — Customer Segmentation

Different customer segments show different patterns of adoption and engagement.

Recommendation

The business should use segment-specific customer strategies rather than applying the same approach to every customer group.

15. Dashboard Features

The Power BI dashboard provides:

- Executive KPI Cards
- Interactive Slicers
- Plan Category Filtering
- Regional Filtering
- Subscription Trend Analysis
- Customer Segmentation
- Feature Adoption Analysis
- Support Issue Analysis
- Business Insights
- Recommendations
- Interactive Navigation
- Drill-down Analysis

16. Dataset Limitations

The following limitations were considered during the project:

- The analysis is based on the available dataset and may not represent the complete SaaS business population.
- Historical data may not fully represent current customer behavior.
- Correlation analysis identifies relationships but does not establish causation.
- Some categories may contain different numbers of observations, which can affect comparisons.
- Dashboard findings depend on the quality and completeness of the source data.
- The project is based on batch data analysis rather than real-time streaming data.

These limitations should be considered when using the results for business decisions.

17. 21-Day Sprint Documentation

Day 1 — Project Initiation

Work Done:

Reviewed the project blueprint, confirmed the SaaS dataset, assessed the source, and created the GitHub repository with the required project folders.

Learning:

Learned how to understand project requirements, evaluate a dataset source, and establish a professional project structure.

Day 2 — Problem & Data Dictionary

Work Done:

Defined the business problem and success metrics, identified the two datasets, mapped the dataset columns to business meanings, and created Data Dictionary v1.

Learning:

Learned how to translate technical fields into business terminology and the importance of maintaining a structured data dictionary.

Day 3 — Data Profiling

Work Done:

Checked row and column counts, data types, missing values, and duplicate records using `record_id` and `customer_id`.

Learning:

Learned how to systematically profile datasets before performing cleaning and analysis.

Day 4 — Data Quality Assessment

Work Done:

Identified numeric-as-text issues, malformed subscription dates, invalid numerical values, and inconsistent categories.

Learning:

Learned how data-quality issues can affect calculations and business analysis.

Day 5 — Cleaning Plan

Work Done:

Prepared a cleaning plan covering missing values, duplicates, type corrections, and category harmonization.

Learning:

Learned how documented cleaning rules improve consistency and reproducibility.

Day 6 — Excel Cleaning

Work Done:

Cleaned the datasets in Excel by correcting types, trimming whitespace, harmonizing categories, and validating row counts.

Learning:

Improved Excel data-cleaning skills and learned the importance of post-cleaning validation.

Day 7 — Excel Analysis

Work Done:

Created pivot tables and analyzed discount, tax, and payment values across business categories.

Learning:

Learned how pivot tables can quickly summarize business information and identify patterns.

Day 8 — SQL Setup

Work Done:

Loaded datasets into SQL, defined keys, reproduced cleaning operations, and created the database schema.

Learning:

Learned how to structure data in a relational database and reproduce cleaning logic using SQL.

Day 9 — SQL Analysis

Work Done:

Created joins, aggregations, and KPI queries across the datasets.

Learning:

Improved SQL skills in joins, grouping, aggregation, and business KPI development.

Day 10 — SQL Views & Window Functions

Work Done:

Created KPI views and implemented ranking, running totals, `LAG`, and `LEAD`.

Learning:

Learned how window functions support advanced time-based and ranking analysis.

Day 11 — Python Cleaning

Work Done:

Reproduced the cleaning process in Pandas and validated row and null counts.

Learning:

Learned how to automate data cleaning and validate results programmatically

Day 12 — EDA

Work Done:

Performed exploratory analysis on discount, tax, and final payment values and analyzed different business segments.

Learning:

Improved skills in Pandas, grouping, aggregation, and exploratory analysis.

Day 13 — Outliers & Visualization

Work Done:

Performed IQR and Z-score outlier analysis and created multiple analytical charts.

Learning:

Learned how to identify unusual observations and select appropriate visualizations.

Day 14 — Descriptive Statistics

Work Done:

Calculated descriptive statistics and analyzed distributions using histograms, skewness, and kurtosis.

Learning:

Improved understanding of statistical distributions and numerical data characteristics.

Day 15 — Statistical Analysis

Work Done:

Created a correlation matrix, performed variance analysis, and statistically validated the main hypothesis.

Learning:

Learned how correlation and variance analysis can support business interpretation.

Day 16 — Power BI Model & KPIs

Work Done:

Imported cleaned data into Power BI, created relationships, developed DAX measures, and created KPI cards.

Learning:

Learned Power BI data modeling and DAX-based KPI development.

Day 17 — Dashboard Development

Work Done:

Created Overview, Subscription Trends, and Customer Segments pages with filters and drill-down functionality.

Learning:

Learned how to structure interactive dashboards and organize visuals around business questions.

Day 18 — Dashboard Design

Work Done:

Improved dashboard titles, annotations, spacing, tooltips, theme consistency, and accessibility.

Learning:

Learned how visual design and storytelling improve dashboard usability.

Day 19 — Insights & Recommendations

Work Done:

Finalized business insights, recommendations, dataset limitations, and cleaning documentation.

Learning:

Learned how to convert analytical findings into actionable business recommendations.

Day 20 — Documentation & Portfolio

Work Done:

Finalized the Data Dictionary and README, documented SQL and Python files, prepared dashboard documentation, and organized portfolio artifacts.

Learning:

Learned how to document an analytics project professionally and prepare it for portfolio presentation.

Day 21 — Final Presentation & Submission

Work Done:

Rehearsed the final business and technical presentation, prepared the technical defence, reviewed the project folder, and prepared the final submission materials.

Learning:

Learned how to present the complete analytics workflow as a business story and confidently explain the technical and analytical decisions.

18. Overall Sprint Learning

The 21-day sprint provided practical experience across the complete data analytics lifecycle.

I gained experience in:

- Data profiling
- Data cleaning
- Excel analysis
- SQL
- SQL joins
- SQL aggregations
- SQL window functions
- Python
- Pandas
- Exploratory Data Analysis
- Descriptive statistics
- Outlier analysis
- Correlation analysis
- Variance analysis
- Power BI
- DAX
- Dashboard development
- Business storytelling
- Documentation
- GitHub
- Project presentation

The project helped me understand how raw business data can be transformed into structured analysis and finally into actionable business insights.

19. Final Project Outcome

The final project successfully combines **data preparation, SQL analysis, Python analysis, statistical validation, and Power BI visualization** into a single business analytics solution.

The Power BI dashboard provides users with an interactive view of:

- Revenue
- Customers
- Subscriptions
- Average payment
- Discounts
- Tax
- Billing cycles
- Product usage

- Customer segments
- Company revenue
- Support issues
- Feature adoption
- Customer satisfaction

The project demonstrates the complete process of transforming raw SaaS data into a business intelligence solution that can support data-driven decision-making.

20. Future Scope

Future improvements can include:

- Real-time SaaS data integration.
- Automated dashboard refresh.
- More advanced customer segmentation.
- Predictive customer behavior analysis.
- Churn prediction using machine learning.
- Customer lifetime value analysis.
- Automated business alerts.
- More detailed regional and plan-level analysis.
- Integration with CRM systems for business action tracking.

21. Project Deliverables

The final project folder contains the required project resources, including:

Documentation

- Overall Project Documentation
- README
- Data Dictionary
- Sprint Documentation
- Data Cleaning Documentation
- Dashboard Documentation
- Business Insights & Recommendations
- Dataset Limitations

Code & Analysis

- SQL scripts
- SQL schema
- SQL queries/views
- Python notebooks
- Data-cleaning scripts
- EDA and statistical analysis

Dashboard

- Power BI .pbix file
- Dashboard screenshots
- Dashboard-related documentation

Presentation

- Final presentation , Demo video

22. Project Links

GitHub Repository

SaaS Product Engagement & Expansion Intelligence

<https://github.com/Rajender1709Reddy/SaaS-Product-Engagement-Expansion-Intelligence>

Power BI Dashboard

https://drive.google.com/file/d/1zMXbf3mnOtPFscQ36V6977QIlzc5dEyq/view?usp=drive_link

Demo Video

https://drive.google.com/file/d/1LplndQdk2SipNwHufhbQFNSfHxePzuEF/view?usp=drive_link

23. Conclusion

The **SaaS Product Engagement & Expansion Intelligence** project successfully demonstrates an end-to-end data analytics workflow.

The project started with understanding and profiling the SaaS datasets, followed by data cleaning and validation using Excel, SQL, and Python. SQL was used for structured data processing and KPI analysis, while Python was used for exploratory and statistical analysis. The processed information was then transformed into an interactive Power BI dashboard.

The final dashboard provides a centralized view of important business metrics, subscription trends, customer segments, product adoption, support issues, and financial performance. The analysis was further converted into business insights and recommendations to support better decision-making. Overall, the project provided practical experience in data analytics, SQL, Python, Pandas, statistics, Power BI, business intelligence, visualization, documentation, and business storytelling, completing the full analytics lifecycle from raw data to business outcome